CSC 4814 & 6814 Efficient AI

Lecture 02

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Assistant Professor, GSU

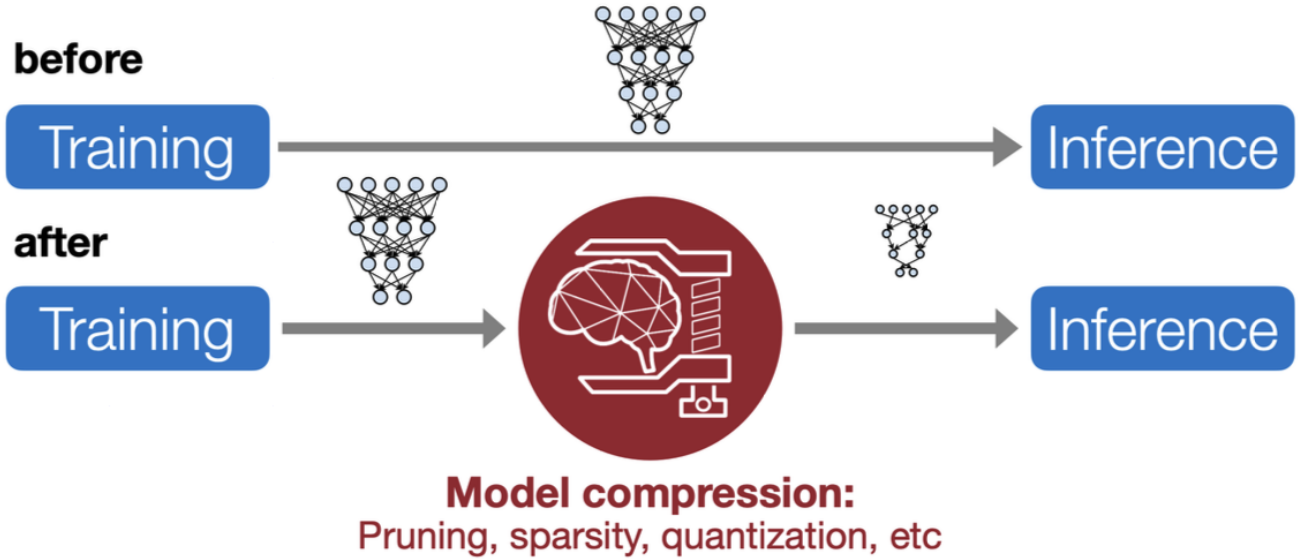
https://www.amai-gsu.us/

haoxinwang@gsu.edu

This course draws inspiration and selected materials from Prof. Song Han's MIT course on TinyML and Efficient Deep Learning Computing. Credit is gratefully acknowledged.

Model Compression and Efficient AI

- **Goal: Doing more with less, making AI fast, small and everywhere**



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Deep Learning is Everywhere

- **But they are computationally costly. How to make them light-weighted and fast?**







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Deep Learning is Everywhere

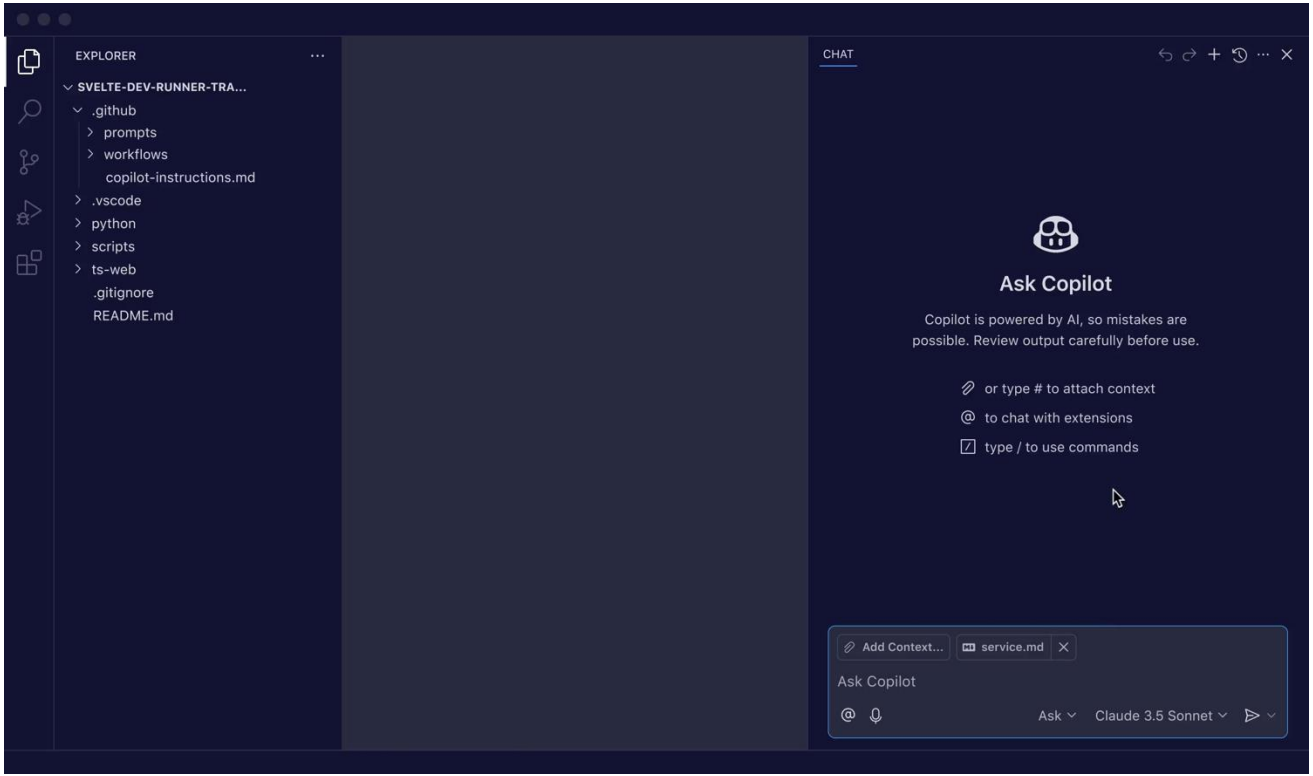
- **But they are computationally costly. How to make them light-weighted and fast?**



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Code Generation

Github CoPilot can make meaningful coding suggestions based on context



https://github.com/features/copilot

Large Language Models

Large language models show emergent behaviors: zero/few learning

Zero-shot

The model predicts the answer given only a natural language description of the task. No gradient updates are performed.

```
1  Translate English to French:        ← task description
2  cheese =>                           ← prompt
```

Few-shot

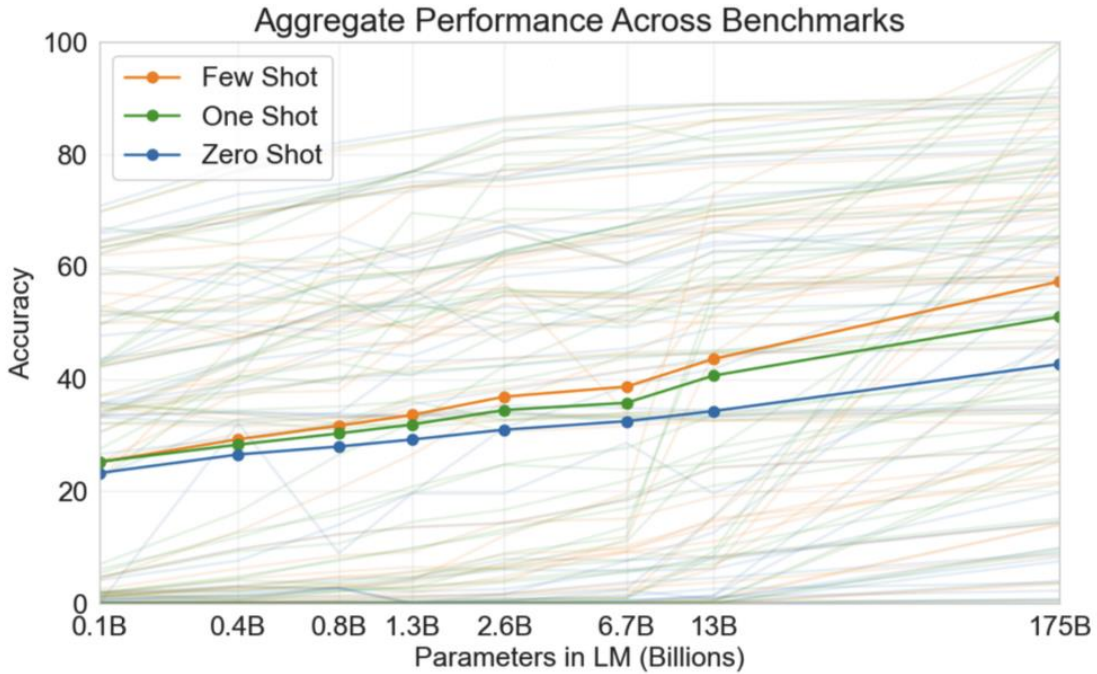
In addition to the task description, the model sees a few examples of the task. No gradient updates are performed.

```
1  Translate English to French:        ← task description
2  sea otter => loutre de mer          ← examples
3  peppermint => menthe poivrée        ←
4  plush girafe => girafe peluche      ←
5  cheese =>                           ← prompt
```



Large Language Models

But it comes at the cost of large model size



Large Language Models

Large language models show emergent behaviors: chain-of-thought

Standard Prompting

Model Input

Q: Roger has 5 tennis balls. He buys 2 more cans of tennis balls. Each can has 3 tennis balls. How many tennis balls does he have now?

A: The answer is 11.

Q: The cafeteria had 23 apples. If they used 20 to make lunch and bought 6 more, how many apples do they have?

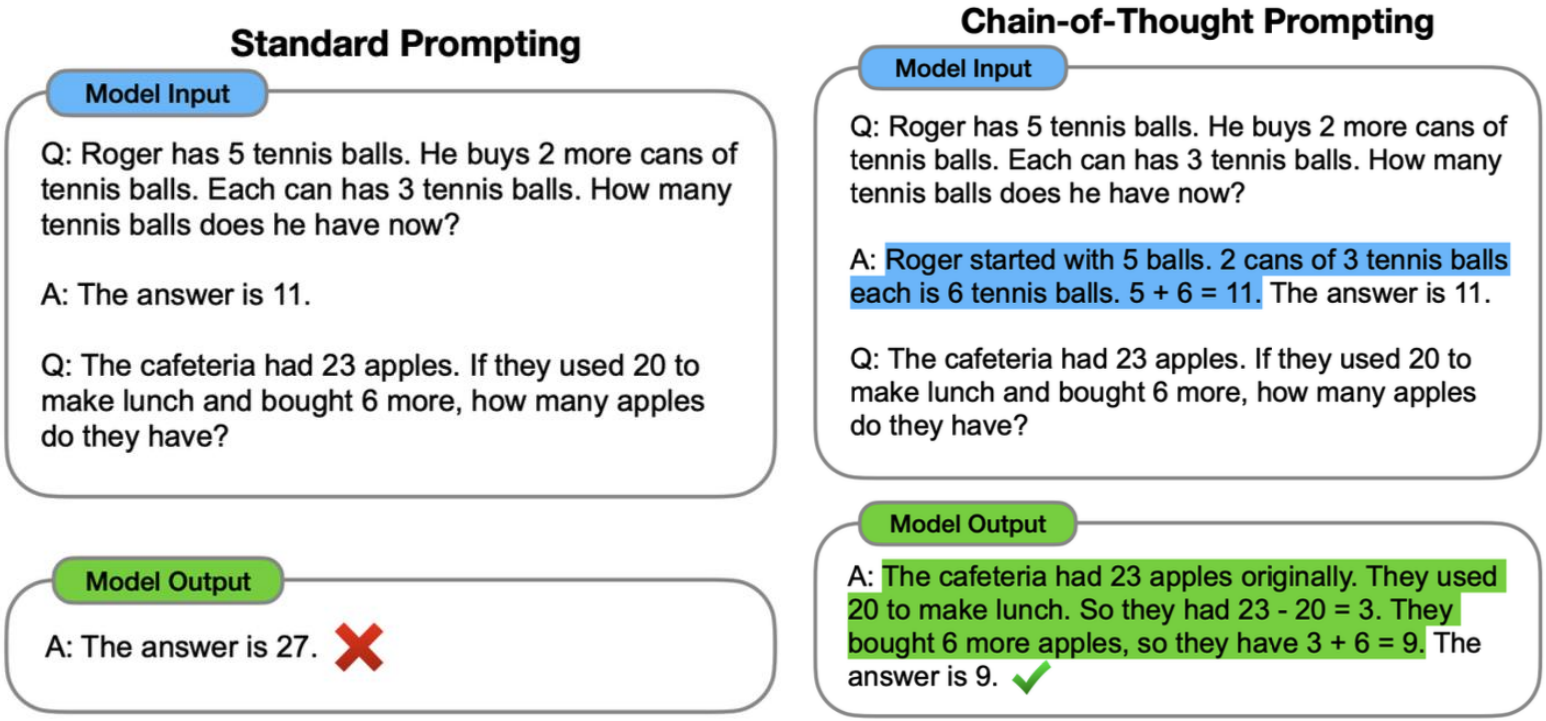
Model Output

A: The answer is 27. ✗

Wei et al., Chain-of-Thought Prompting Elicits Reasoning in Large Language Models, 2022

Large Language Models

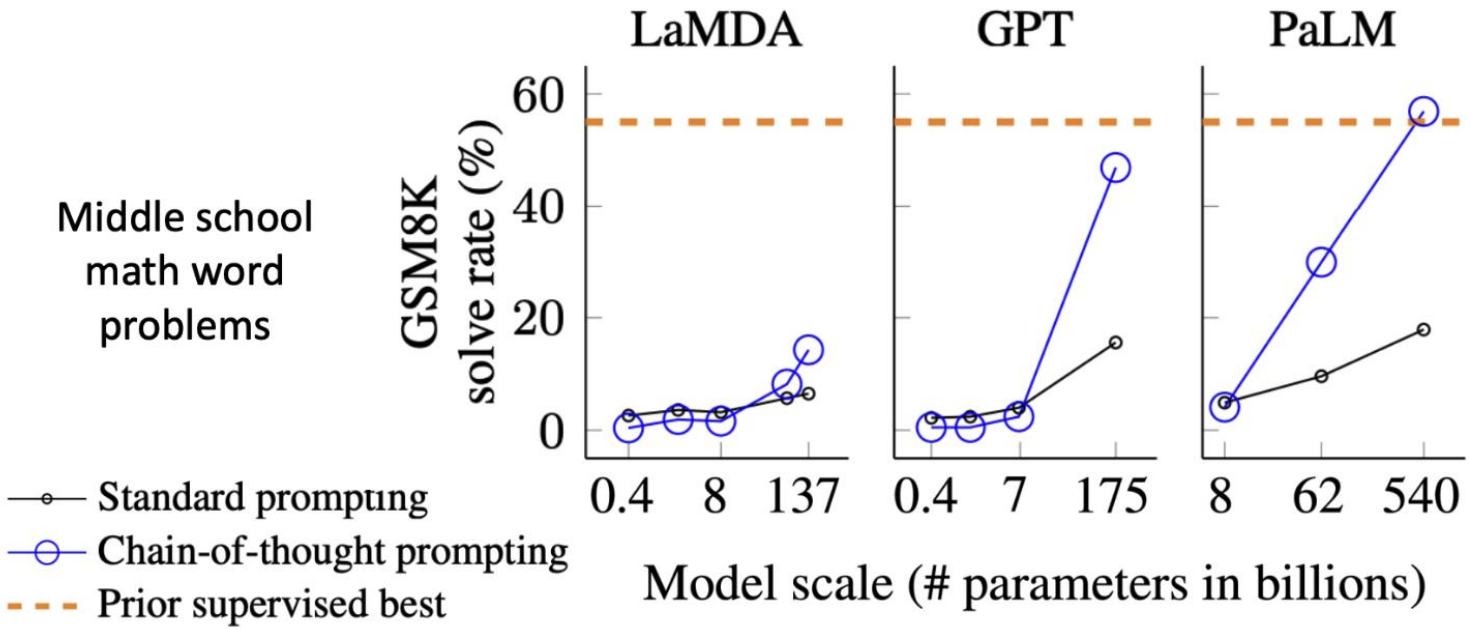
Large language models show emergent behaviors: chain-of-thought



Wei et al., Chain-of-Thought Prompting Elicits Reasoning in Large Language Models, 2022

Large Language Models

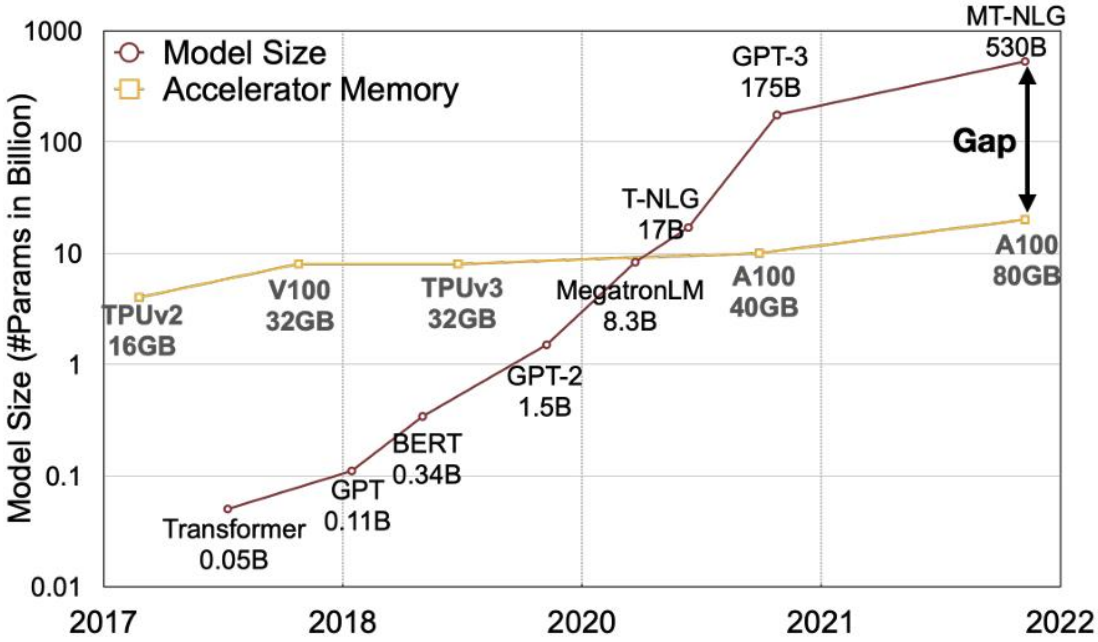
But it comes at the cost of large model size



Wei et al., Chain-of-Thought Prompting Elicits Reasoning in Large Language Models, 2022

Large Language Models

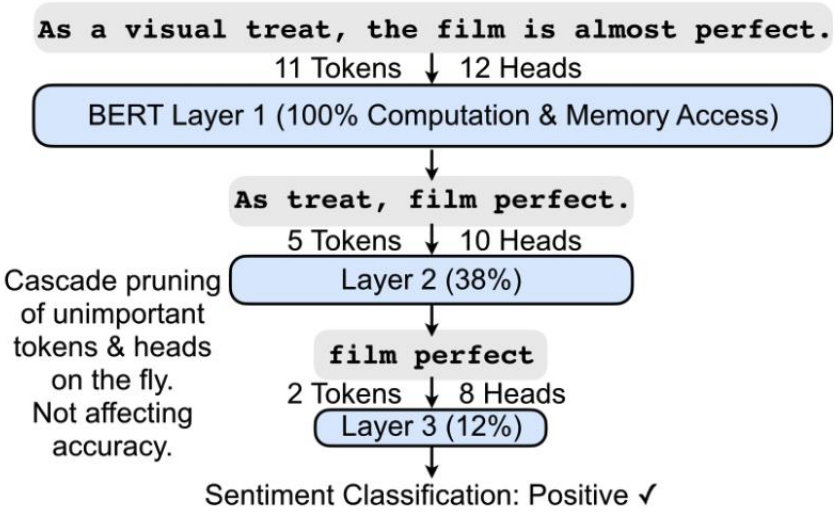
But it comes at the cost of large model size

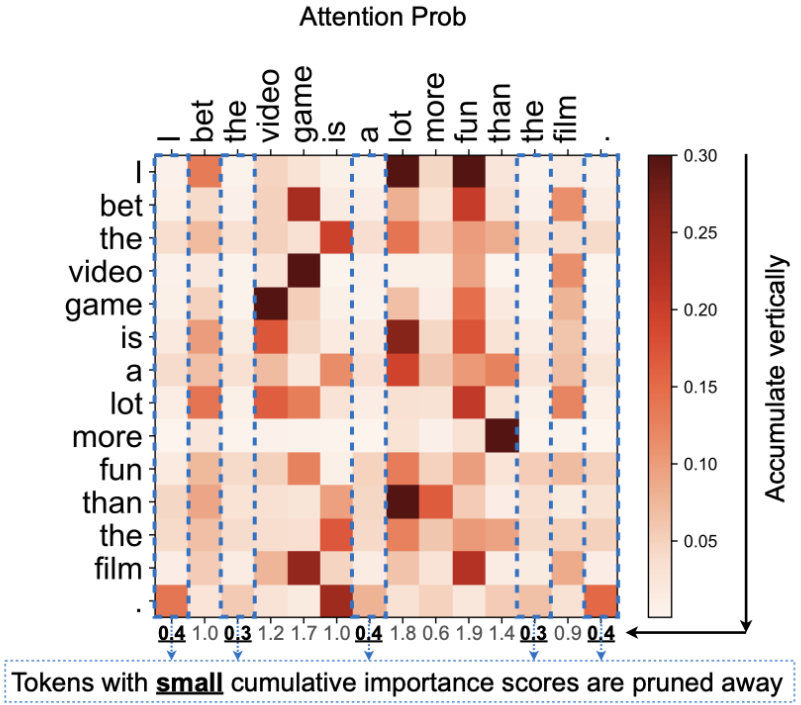


SmoothQuant: Accurate and Efficient Post-Training Quantization for Large Language Models (ICML 2023)

Efficient Large Language Models

- **SpAtten accelerates language models by pruning redundant tokens**

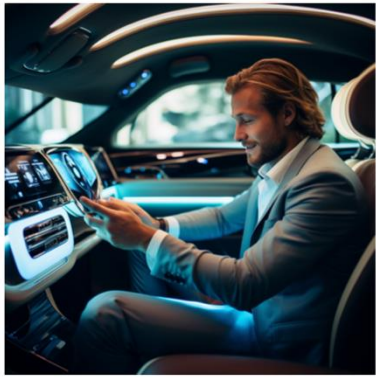
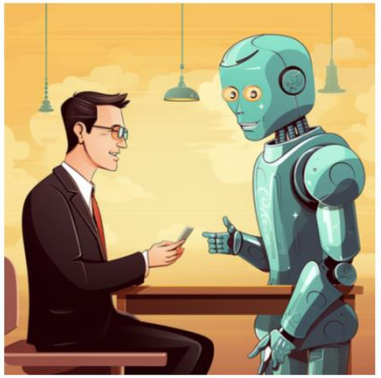




SpAtten: Efficient Natural Language Processing [Wang *et al.*, HPCA 2021]

Efficient Large Language Models

- **Running LLMs on the edge is very important**

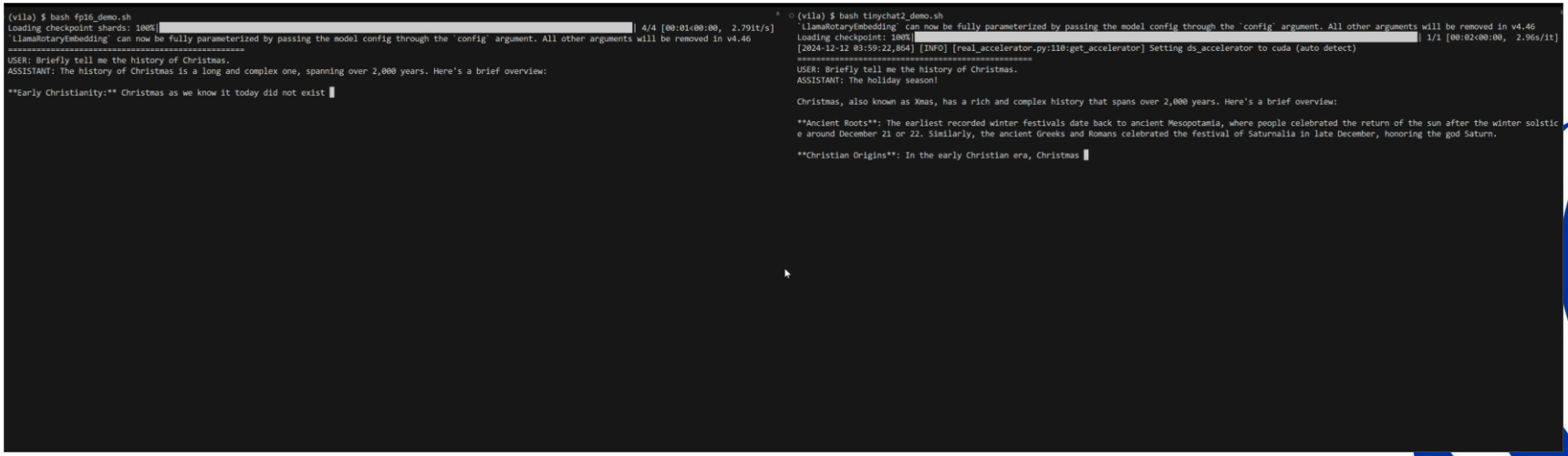





- Deploying LLM on the edge is useful: running copilot services (code completion, office, game chat) locally on laptops, cars, robots, and more. These devices are **resource-constrained**, **low-power** and sometimes **do not have access to the Internet**.

- **Data privacy** is important. Users do not want to upload personal data to the cloud.

Activation-aware Weight Quantization (AWQ) for LLM

AWQ brings about 3.3x speedup to LLaMA-2 on 4090



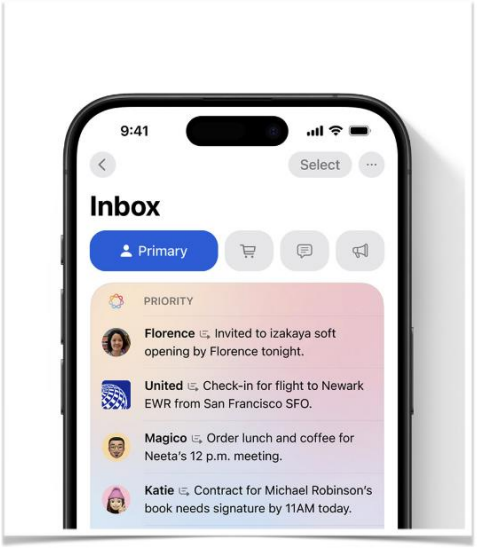
LLaMA-2-7B (FP16): 50 tokens / s

LLaMA-2-7B (W4A16, AWQ): 166 tokens / s

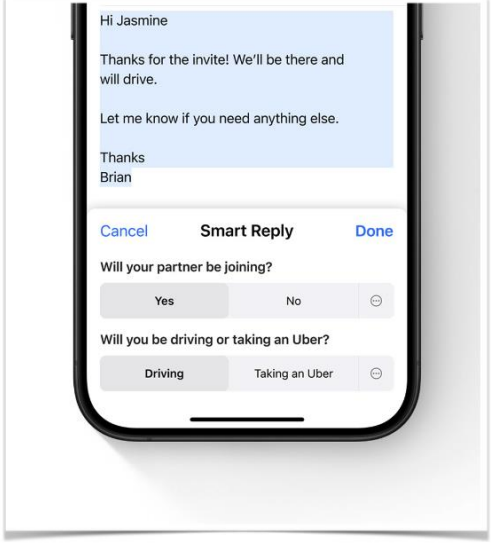
https://github.com/mit-han-lab/llm-awq

LLMs on Phones

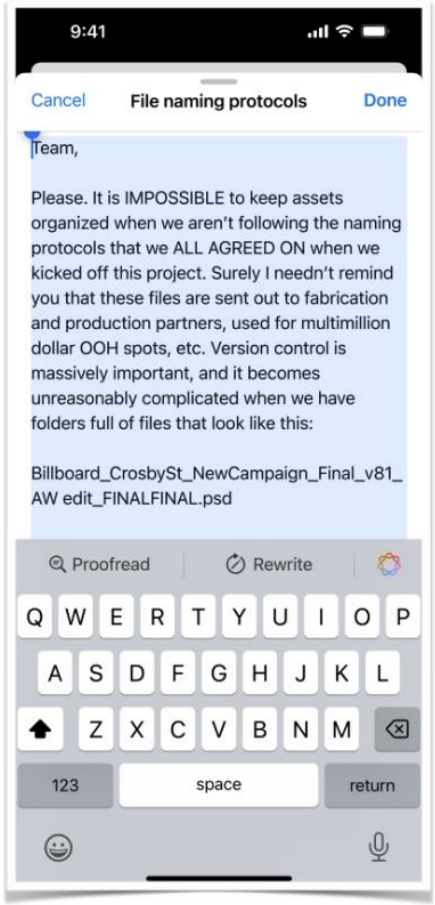
- LLM empowers writing tools on phones



Summarization



Smart Reply



Writing Tools

https://www.apple.com/apple-intelligence/

Deep Learning is Everywhere

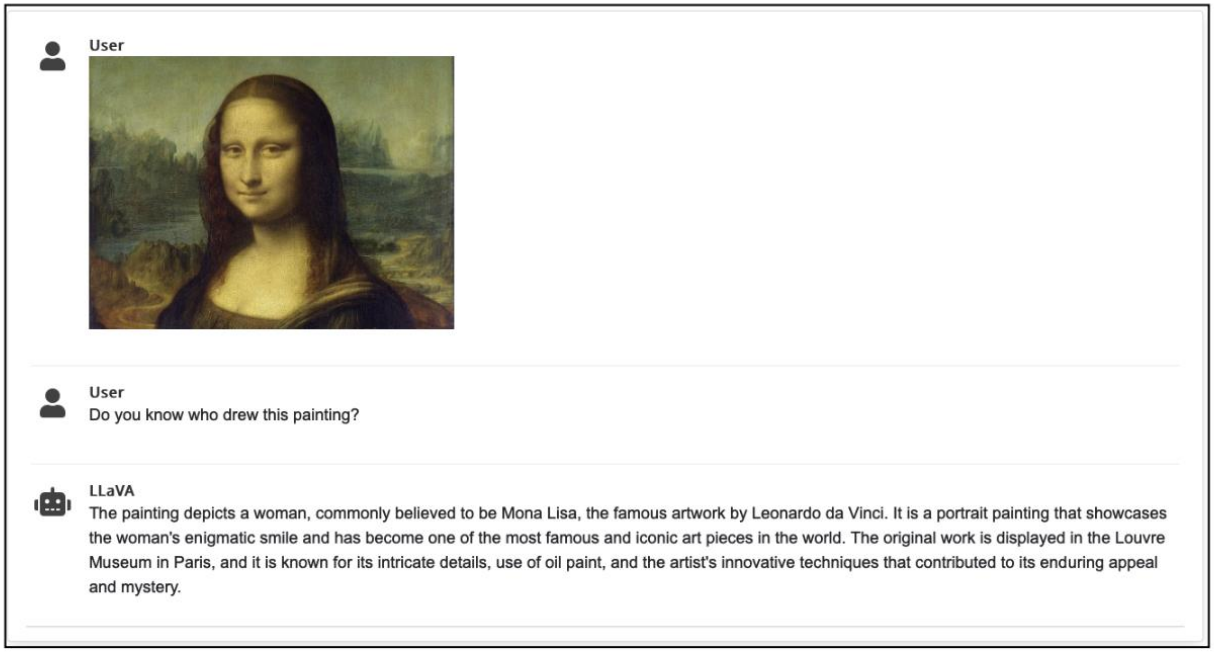
- **But they are computationally costly. How to make them light-weighted and fast?**



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Vision-Language Models

LLaVA achieves general-purpose visual and language understanding



LLaVA uses a 13B LLaMA for language understanding.

https://llava-vl.github.io/

Efficient Vision-Language Models

AWQ quantizes vision-language models to 4 bits with high quality



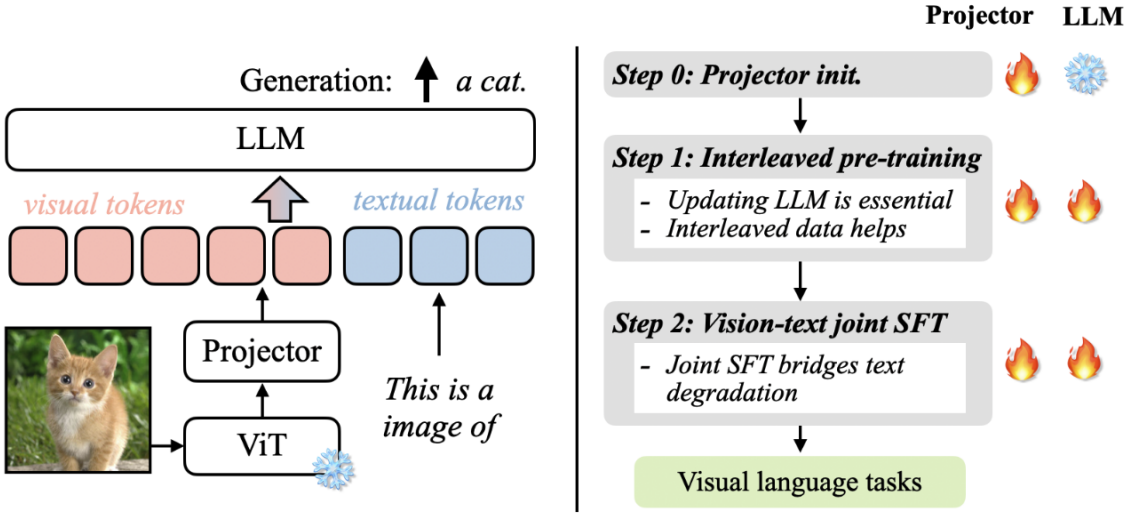
Question: Do you know who drew this painting?
INT4-g128 RTN: The painting depicts a woman, who some believe to be a representation of the famous artist, Mona Lisa. However, it is not certain that the woman in the painting is indeed Mona Lisa. The artwork is a popular subject of artistic analysis and admiration, often considered a symbol of feminine beauty and enigma. The portrait is characterized by its intricate details, captivating the viewer with the woman's expressive gaze and the subtle, enigmatic smile that has become synonymous with the Mona Lisa.
INT4-g128 AWQ: The painting depicts a woman, who is believed to be a portrait of Mona Lisa, the famous artwork by the artist Leonardo da Vinci. Although the painting is widely recognized as a representation of Mona Lisa, it is not possible to confirm the artist who drew this particular painting without additional information. Leonardo da Vinci was a renowned artist known for his exceptional skill in painting detailed, realistic portraits, and the Mona Lisa is one of his most famous works.



Question: Can you explain this meme in detail?
INT4-g128 RTN: The meme in the image features a plate of food, likely fried chicken, with a unique twist. There are small pictures of the Earth and other planets placed on top of the food, creating an interesting and amusing visual. Alongside the plate, a caption reads, "Sometimes I look at pictures of the Earth from space and I marvel at how beautiful it all is." This combination of food and planet images adds a touch of humor and curiosity, making the viewer wonder about the context and intent behind the meme.
INT4-g128 AWQ: The meme in the image is a lighthearted and humorous take on the concept of looking at pictures of the Earth from space. A plate of fried food, specifically chicken nuggets, is presented with the caption, "Sometimes I look at pictures of the Earth from space and I marvel at how beautiful it all is." The playful combination of a food item and an Earth-related context creates a fun and entertaining visual.

AWQ: Activation-aware Weight Quantization for LLM Compression and Acceleration [Lin *et al.*, 2023]

Vision-Language Models

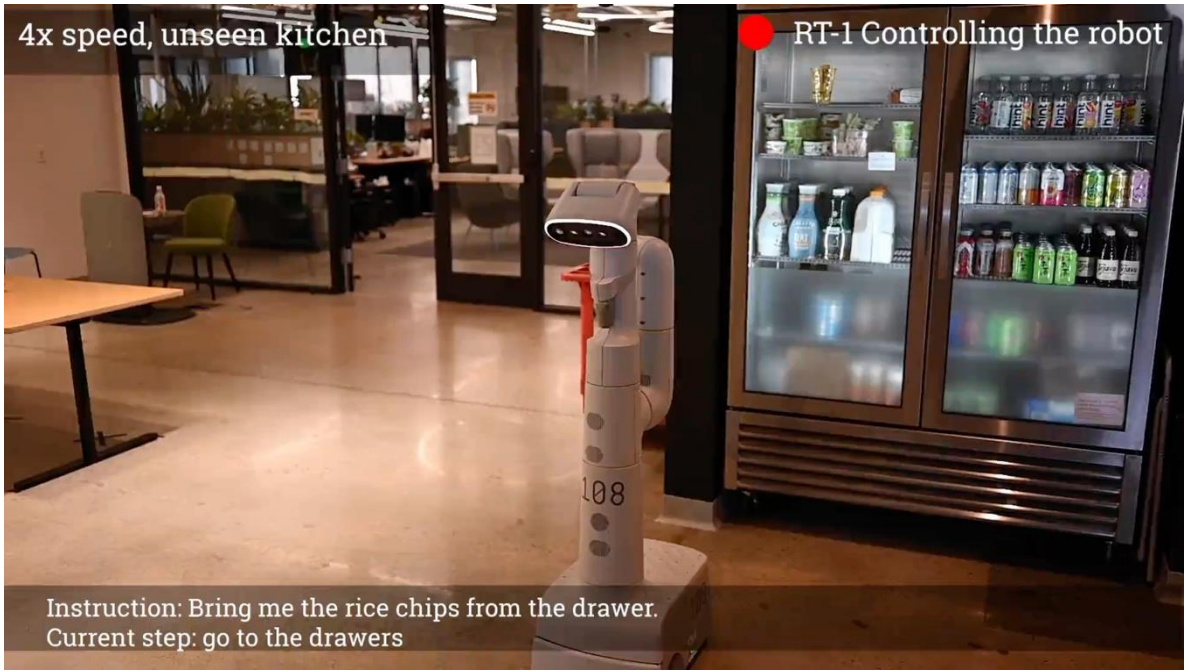


- Data and training recipe matters more than architecture.
- Visual instruction tuning is not enough; we need visual language pre-training.
- Image-text pairs are not enough; interleaved image text data is crucial.
- Visual in-context learning requires unfreezing the LLM.
- Re-blending text-only data recovers text-only capability

Vision-Language-Action Models

- **Robotics transformers control robots based on language instructions**

Robotics at Google · Everyday Robots · Google Research



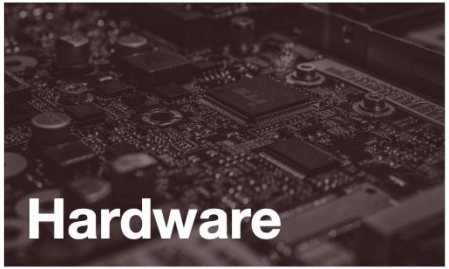
"Bring me the rice chips from the drawer" is executed in an office kitchen that the robot has never seen before.

https://robotics-transformer1.github.io/

Run at only 3Hz due to the high computational cost and networking latency.

Deep Learning: Three Pillars









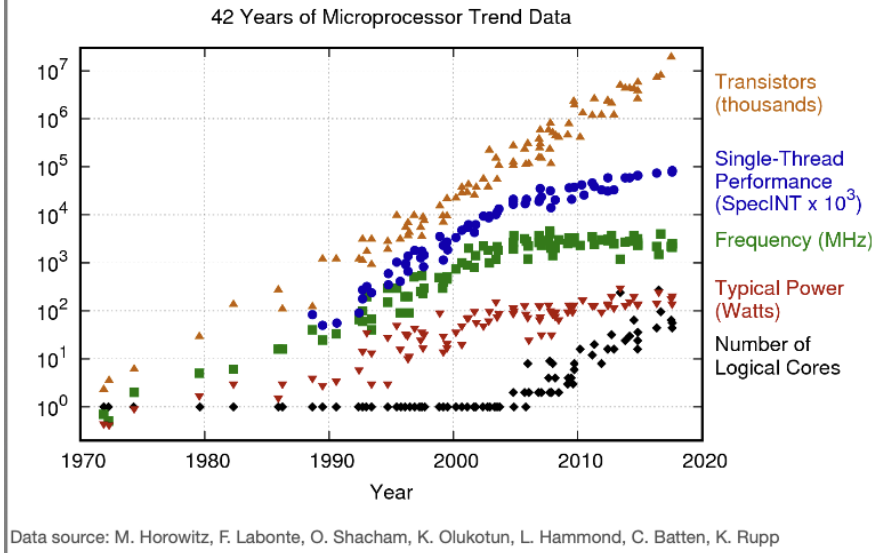




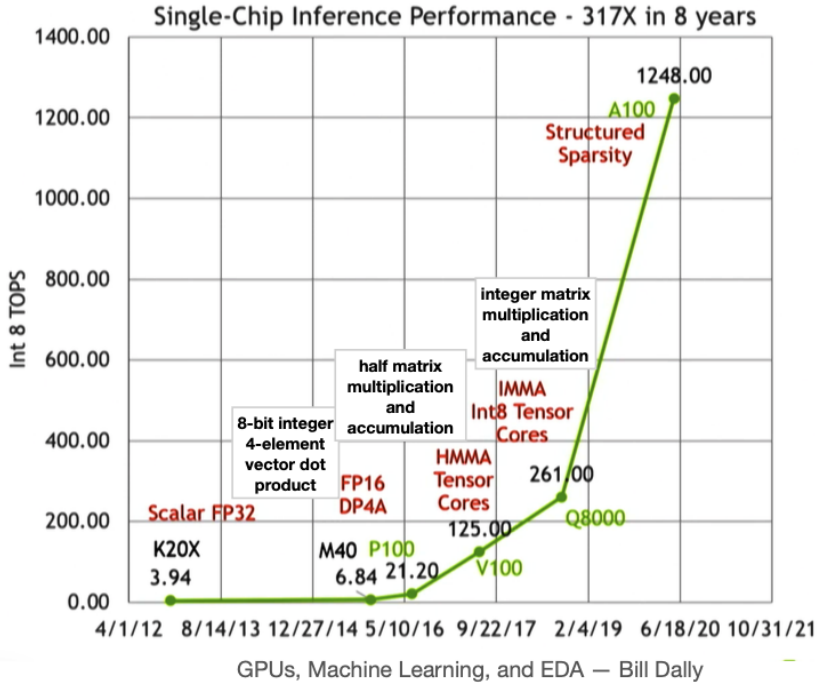
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Architectural support for quantization/pruning brings tremendous improvement

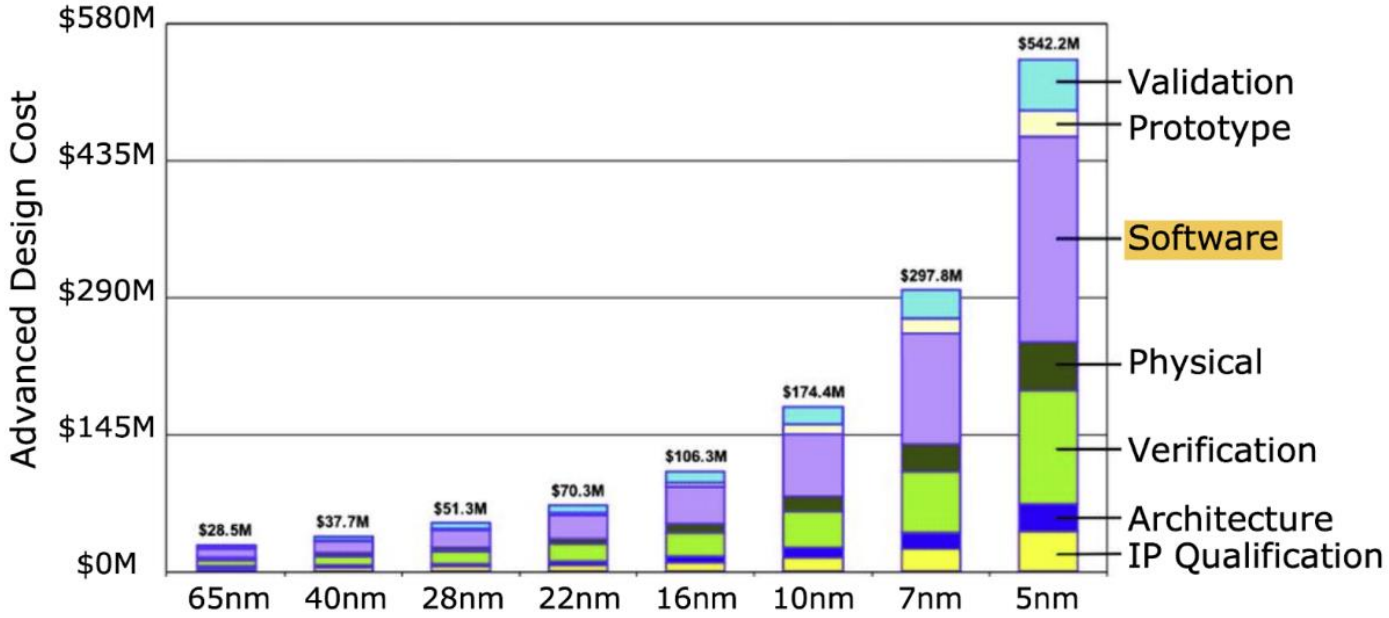
FP32 => FP16 => Int8; dense => sparse



Data source: M. Horowitz, F. Labonte, O. Shacham, K. Olukotun, L. Hammond, C. Batten, K. Rupp



Software Innovation is important in advanced technology node



The software cost dominates the cost breakdown of advanced technology nodes [source].
We focus on designing new algorithms and software for efficient computing.

Cloud AI Hardware

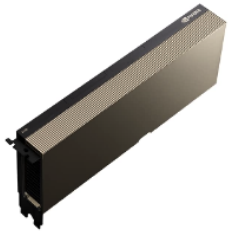
NVIDIA



P100 (2016)



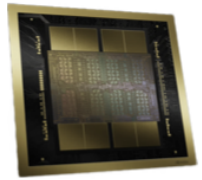
V100 (2017)



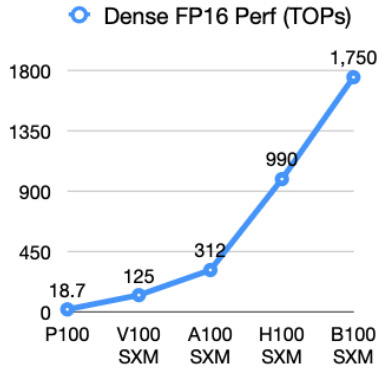
A100 (2020)

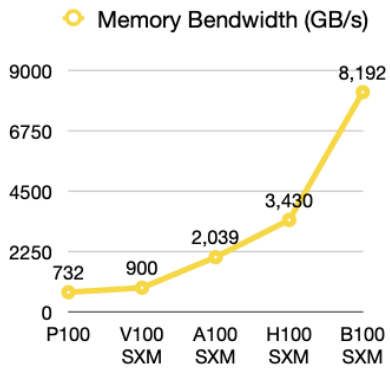


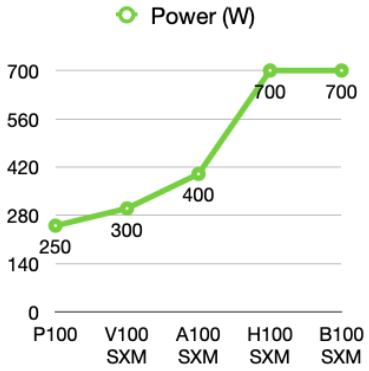
H100 (2022)

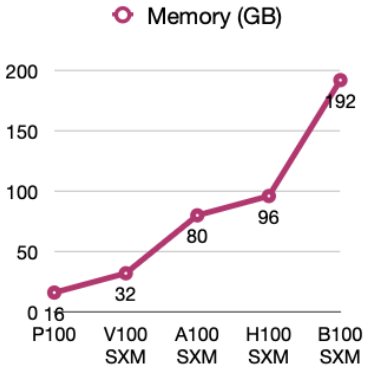


B100 (2024)









Edge AI Hardware

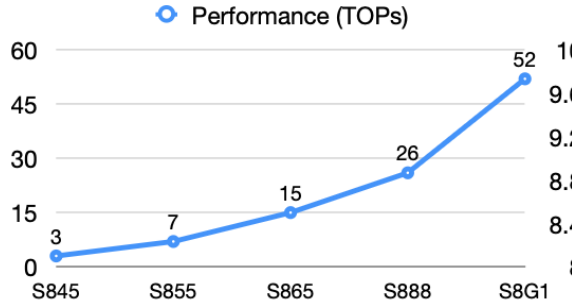
Qualcomm Hexagon DSP

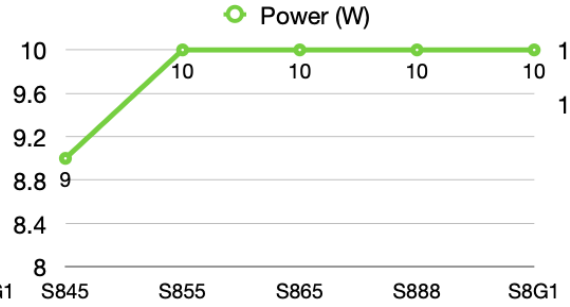
- Qualcomm Hexagon is a family of digital signal processor (DSP) products by Qualcomm. It is designed to deliver performance with low power over a variety of applications

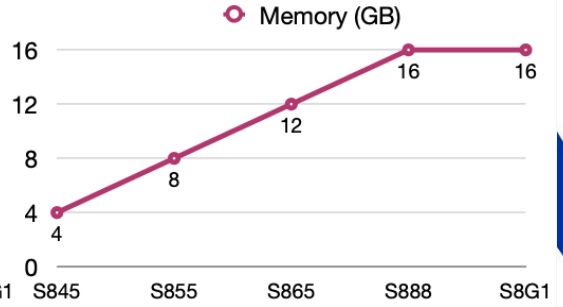












Edge AI Hardware

Apple Neural Engine

- The Apple Neural Engine (ANE) is an energy-efficient and high-throughput engine for ML inference on Apple silicon.



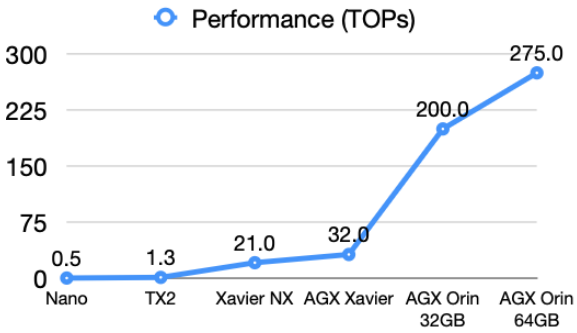
NanoReview. https://nanoreview.net/en

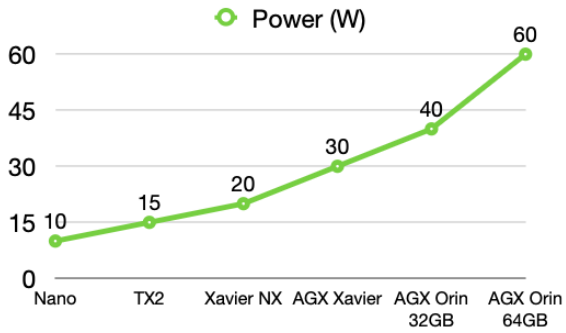
Edge AI Hardware

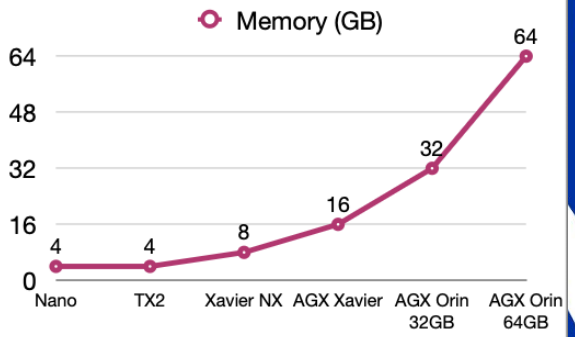
Nvidia Jetson

- NVIDIA Jetson is a complete System on Module (SOM) that includes a GPU, CPU, memory, power management, high-speed interfaces, and more









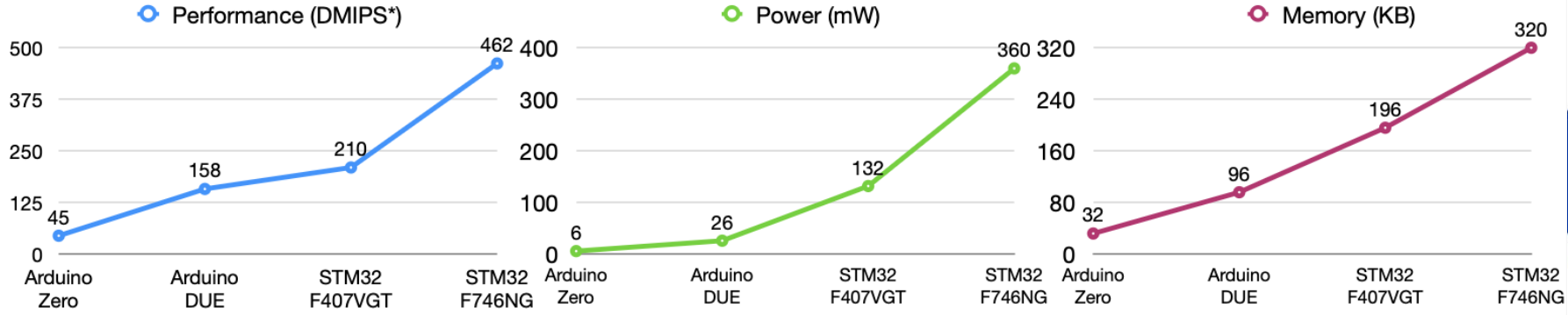
NanoReview. https://connecttech.com/jetson/jetson-module-comparison/

Edge AI Hardware

Microcontrollers (MCU)

- A microcontroller is a compact integrated circuit designed for embedded systems. A typical microcontroller includes a processor, memory and input/output (I/O) peripherals on a single chip.





* Dhrystone Million Instructions Per Second (DMIPs) is an index for integer computation.

Edge AI Hardware

- Edge AI devices still have huge gap to cloud processors



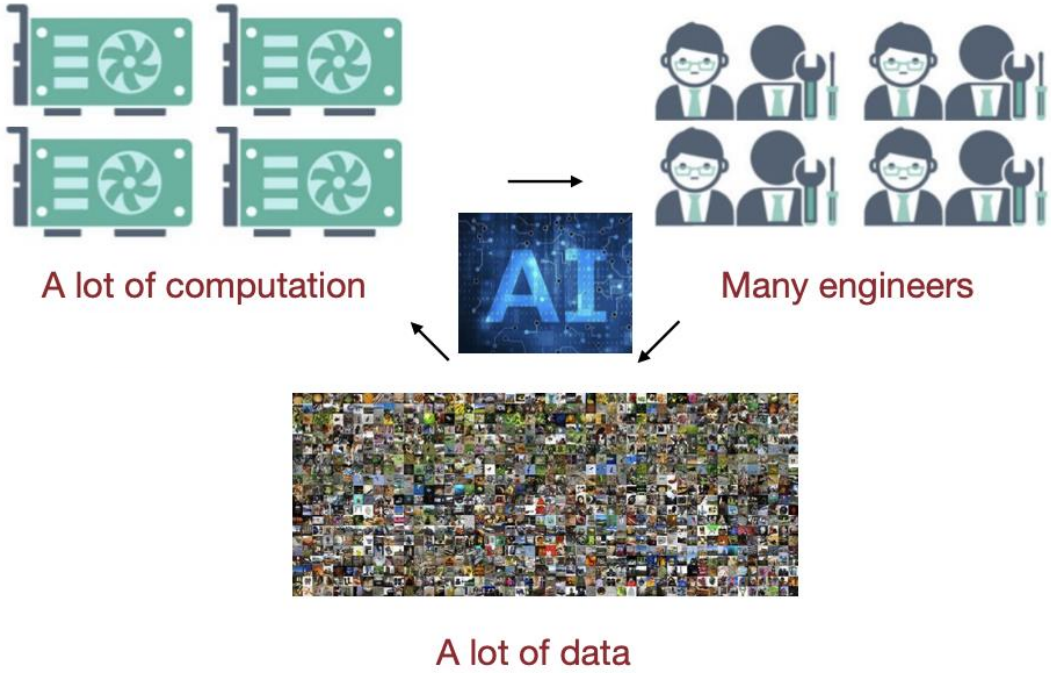




	Cloud AI	Mobile AI	Tiny AI
Memory (Activation)	80GB	4GB	320kB
Storage (Weights)	~TB/PB	256GB	1MB

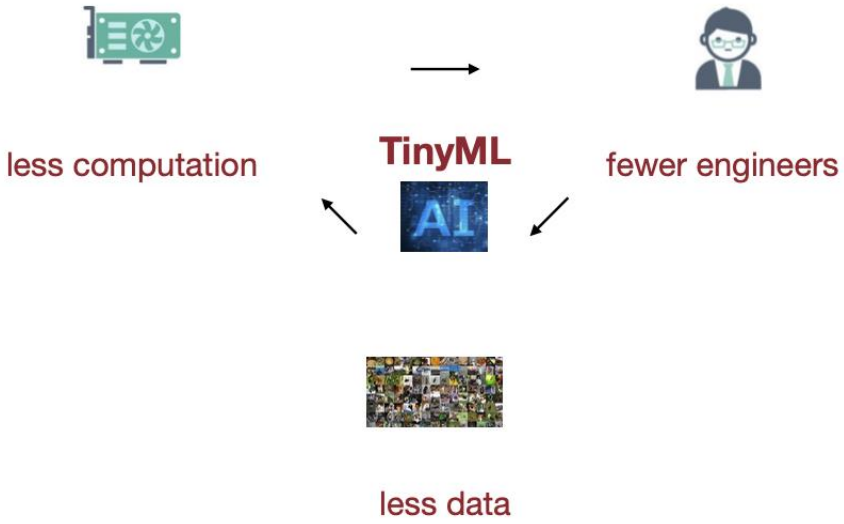
Current Landscape of Deep Learning

- Big computation, engineer and data

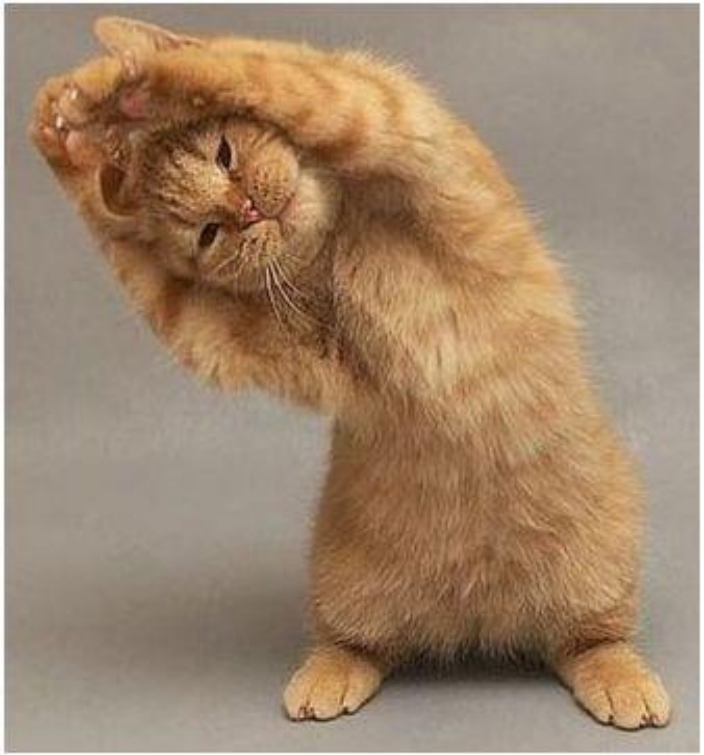


Tiny Deep Learning

- Tiny computation, engineer and data



BREAK

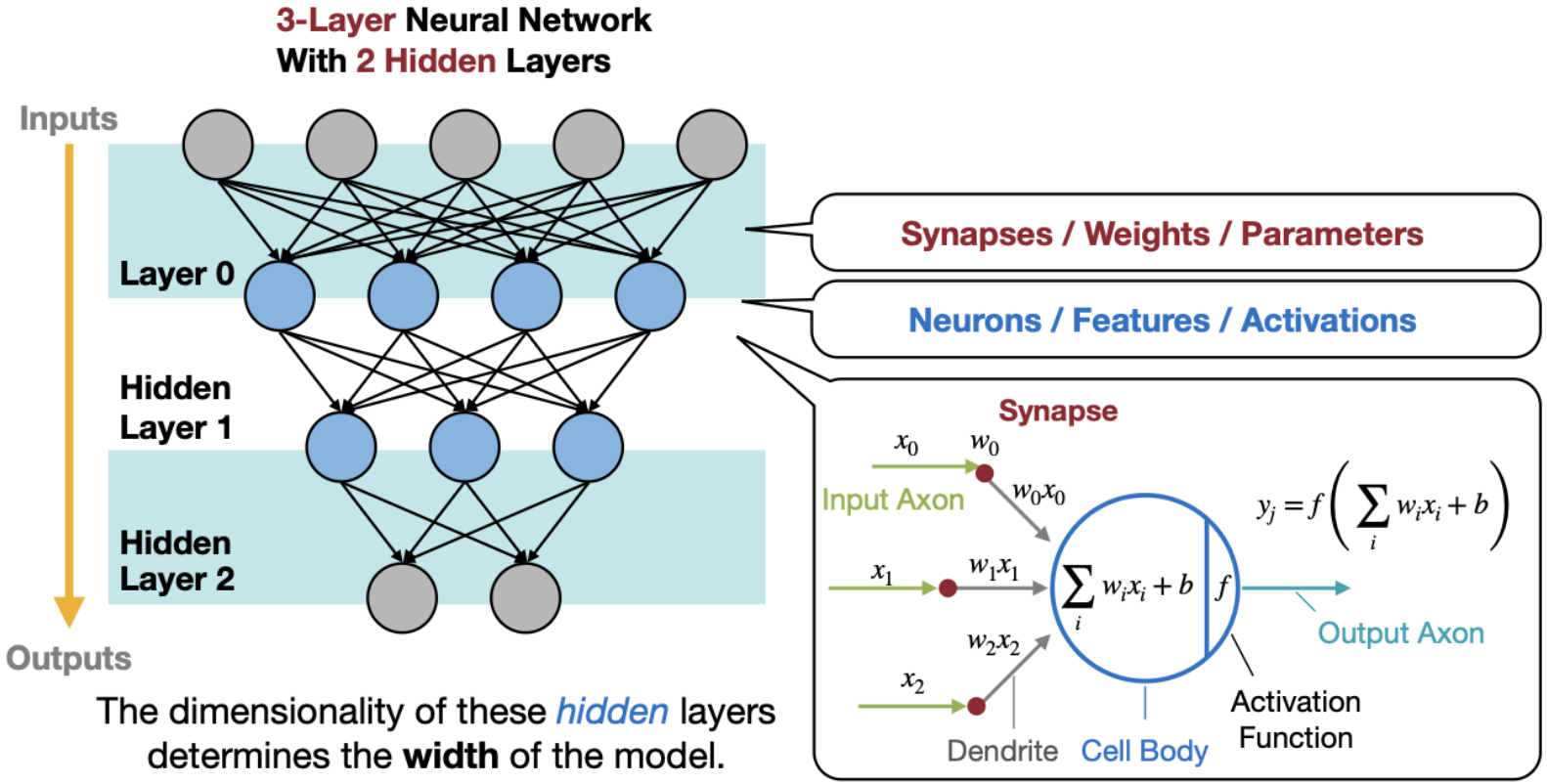


Plan for Lectures 2-3

1. Review the **terminology of neural networks**
   - Neuron, Synapses, Activation, Feature, Weight, Parameter, *etc*
2. Review **popular building blocks** in a neural network
   - Fully-Connected, Convolution, Grouped Convolution, Depthwise Convolution
   - Pooling, Normalization, Transformer
3. Introduce popular efficiency metrics for neural networks
   - #Parameters, Model Size, Peak #Activations, MAC, FLOP, FLOPS, Latency, Throughput



Deep Neural Network



The dimensionality of these *hidden* layers determines the **width** of the model.

Popular Neural Network Layers



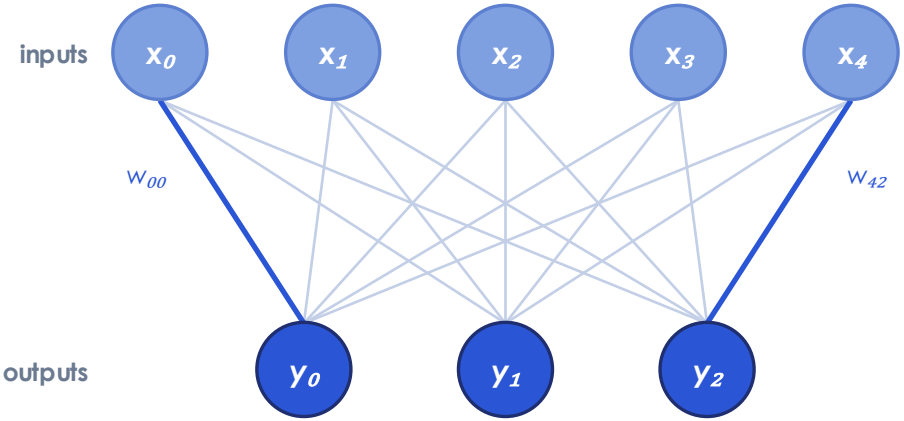
Fully-Connected Layer

Every output neuron connects to every input neuron

The layer that starts it all: no connection is skipped. Once you see it as a single matrix multiply, the whole thing collapses into one line of math.

$$Y = XW^T + b$$

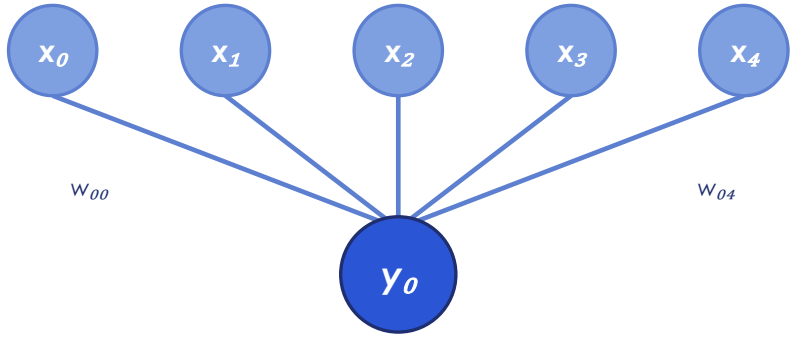
one matrix multiply, plus a bias



One Neuron = a Weighted Sum

Give each input a weight, add them up, add a bias

Intuition. Think of x as 5 features of a person (height, weight, age…). To score "fits a basketball team," weight each feature by how much it matters, sum them, then shift by a bias baseline.



$$y_0 = w_{00} \cdot x_0 + w_{01} \cdot x_1 + w_{02} \cdot x_2 + w_{03} \cdot x_3 + w_{04} \cdot x_4 + b_0$$

In general, for every output i: $y_i = \Sigma_j \, w_{ij} \cdot x_j + b_i$

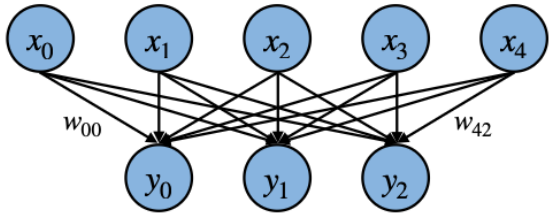
the Σ just compresses the line above

Fully-Connected Layer (Linear layer)

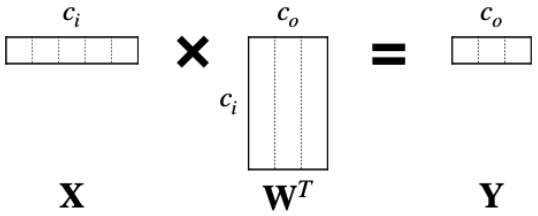
The output neuron is connected to all input neurons.

- **Shape of Tensors:**
  - Input Features $\mathbf{X}$: $(1, c_i)$
  - Output Features $\mathbf{Y}$: $(1, c_o)$
  - Weights $\mathbf{W}$: (c_o, c_i)
  - Bias $\mathbf{b}$: $(c_o,)$

Notations	
c_i	Input Channels
c_o	Output Channels



$$y_i = \sum_j w_{ij} x_j + b_i$$

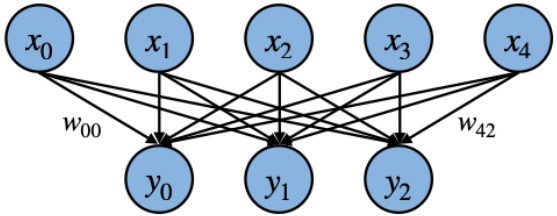


Fully-Connected Layer (Linear layer)

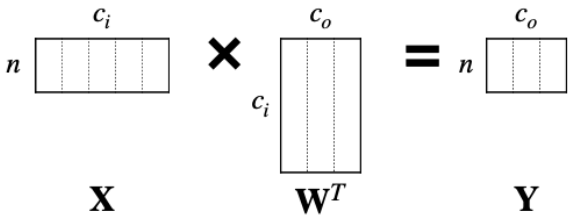
The output neuron is connected to all input neurons.

- **Shape of Tensors:**
  - Input Features $\mathbf{X}$: (n, c_i)
  - Output Features $\mathbf{Y}$: (n, c_o)
  - Weights $\mathbf{W}$: (c_o, c_i)
  - Bias $\mathbf{b}$: $(c_o,)$

Notations	
n	Batch Size
c_i	Input Channels
c_o	Output Channels



$$y_i = \sum_j w_{ij} x_j + b_i$$

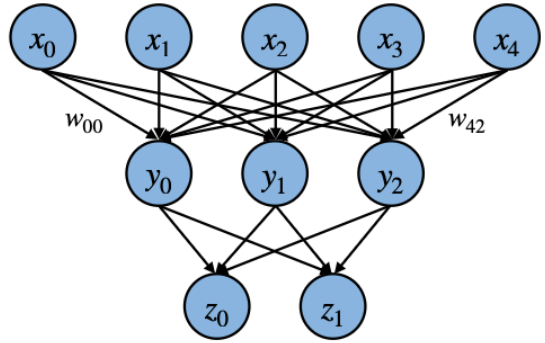


Fully-Connected Layer (Linear layer)

The output neuron is connected to all input neurons.

- **Shape of Tensors:**
  - Input Features $\mathbf{X}$: (n, c_i)
  - Output Features $\mathbf{Y}$: (n, c_o)
  - Weights $\mathbf{W}$: (c_o, c_i)
  - Bias $\mathbf{b}$: $(c_o,)$

Notations	
n	Batch Size
c_i	Input Channels
c_o	Output Channels



Multilayer Perceptron (MLP)

